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GRADE A+
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Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 4

Title: Cleaning Data using Power Query

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Data Cleaning

GitHub Repository Link: [rackxy/Business_analytics](https://github.com/rackxy/Business_analytics)

Problem Statement

A company has received customer data containing missing values, duplicate records, and inconsistent entries. Clean and prepare the dataset using Power Query before analysis.

Dataset used: Cafe Sales Dirty Data

Source: <https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training>

Steps:

Step 1 : Data Import

The customer dataset was imported into Power Query using Microsoft Excel/Power BI.

The data was reviewed to understand the available columns, values, and existing data-quality issues.

File path: C:\Users\hngag\Downloads\archive (4)\dirty_cafe_sales.csv

File origin: 65001: Unicode (UTF-8) | Delimiter: Comma | Data type detection: Based on first 200 rows

Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8
Transaction ID	Item	Quantity	Price Per Unit	Total Spent	Payment Method	Location	Transaction Date
TXN_1961373	Coffee	2	2.0	4.0	Credit Card	Takeaway	2023-09-08
TXN_4977031	Cake	4	3.0	12.0	Cash	In-store	2023-05-16
TXN_4271903	Cookie	4	1.0	ERROR	Credit Card	In-store	2023-07-19
TXN_7034554	Salad	2	5.0	10.0	UNKNOWN	UNKNOWN	2023-04-27
TXN_3160411	Coffee	2	2.0	4.0	Digital Wallet	In-store	2023-06-11
TXN_2602893	Smoothie	5	4.0	20.0	Credit Card		2023-03-31
TXN_4433211	UNKNOWN	3	3.0	9.0	ERROR	Takeaway	2023-10-06
TXN_6699534	Sandwich	4	4.0	16.0	Cash	UNKNOWN	2023-10-28
TXN_4717867		5	3.0	15.0		Takeaway	2023-07-28
TXN_2064365	Sandwich	5	4.0	20.0		In-store	2023-12-31
TXN_2548360	Salad	5	5.0	25.0	Cash	Takeaway	2023-11-07
TXN_3051279	Sandwich	2	4.0	8.0	Credit Card	Takeaway	ERROR
TXN_7619095	Sandwich	2	4.0	8.0	Cash	In-store	2023-05-03
TXN_9437049	Cookie	5	1.0	5.0		Takeaway	2023-06-01
TXN_8915701	ERROR	2	1.5	3.0		In-store	2023-03-21
TXN_2847255	Salad	3	5.0	15.0	Credit Card	In-store	2023-11-15
TXN_3765707	Sandwich	1	4.0	4.0			2023-06-10
TXN_6769710	Juice	2	3.0	6.0	Cash	In-store	2023-02-24
TXN_8876618	Cake	5	3.0	15.0	Cash	ERROR	2023-03-25

Step 2: Data Cleaning

Step 1: Import the Dataset

The customer dataset was imported into Power Query using Microsoft Excel/Power BI.

The data was reviewed to understand the available columns, values, and existing data-quality issues.

Step 2: Examine the Dataset

The dataset was inspected using Power Query's data preview and column-quality features.

The following issues were identified:

- Missing or blank values.
- Duplicate customer records.
- Inconsistent spelling and formatting.
- Different representations of similar values.
- Incorrect or inappropriate data types.

Step 3: Handle Missing Values

Missing values were identified in the relevant columns.

Depending on the nature of the column, missing values were handled appropriately.

Unnecessary blank records were removed, while meaningful missing information was either replaced with a suitable value or retained where appropriate.

This helped reduce incomplete records and improved the reliability of the dataset.

Step 4: Remove Duplicate Records

Duplicate customer records were identified using relevant customer-identifying fields.

Power Query's Remove Duplicates option was used to eliminate repeated records.

This ensured that the same customer was not counted multiple times during analysis.

Step 5: Standardize Inconsistent Entries

Inconsistent values were identified and standardized.

For example, the same category or customer attribute may have appeared in different formats due to differences in spelling, capitalization, or spacing.

Power Query transformations such as Replace Values, Format, Trim, and Clean were used to make the entries consistent.

Step 6: Correct Data Types

The data types of the columns were reviewed and corrected.

For example:

- Customer IDs were treated as appropriate identifier fields.
- Names and categories were stored as text.
- Age and numerical values were converted to numeric data types.
- Dates were converted to date format.
- Financial values were converted to appropriate numeric/currency formats.

Correct data types are important because they ensure that calculations and

visualizations work correctly.

Step 7: Clean Text Data

Text fields were cleaned to remove unnecessary spaces and unwanted characters.

The Trim and Clean transformations were used where required.

Capitalization and formatting were also standardized to improve consistency.

Step 8: Validate the Cleaned Data

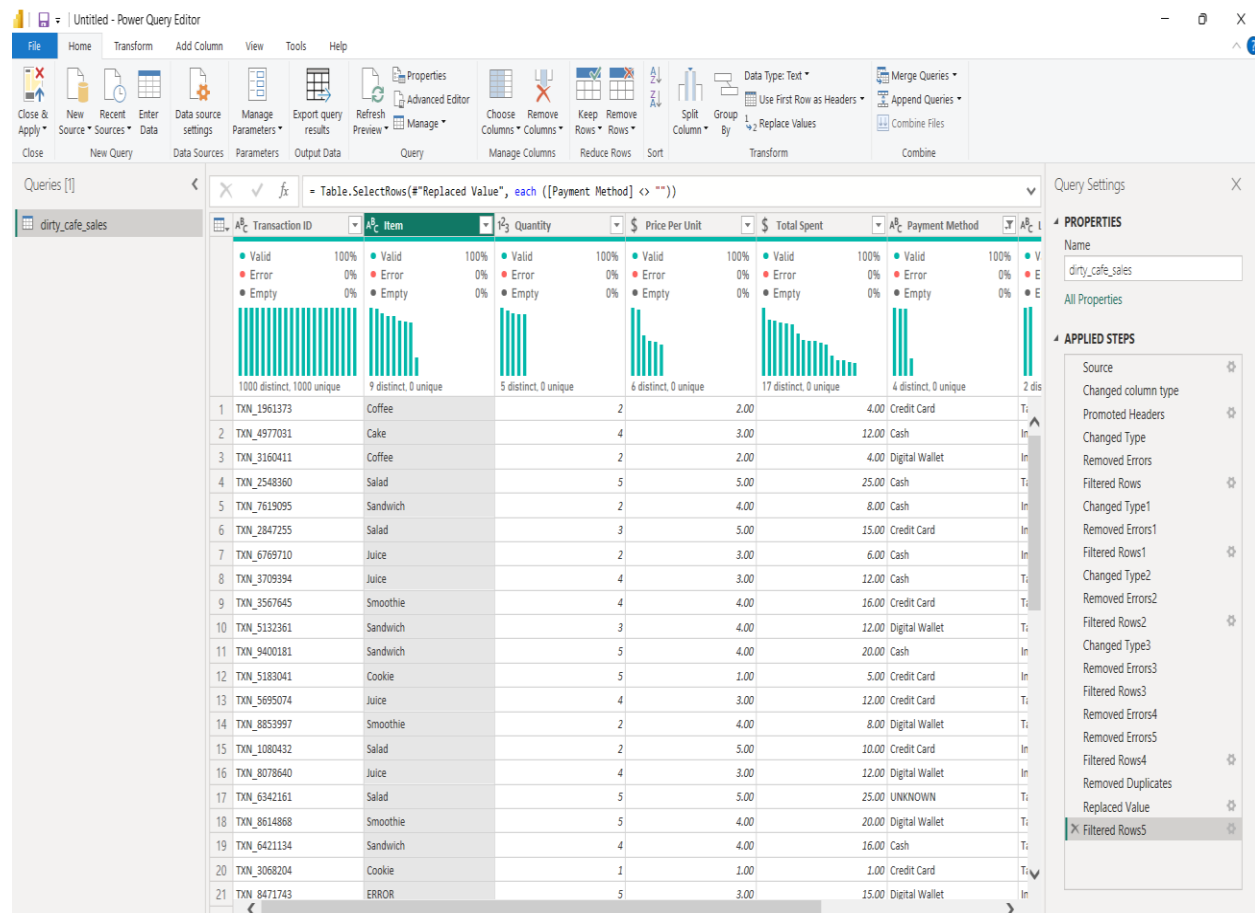
After applying the transformations, the dataset was reviewed again to confirm that:

- Duplicate records had been removed.
- Missing values had been appropriately handled.
- Inconsistent entries had been standardized.
- Data types were correct.
- The resulting dataset was suitable for analysis.

Step 9: Load the Cleaned Dataset

After completing the cleaning and transformation process, the final dataset was loaded back into Excel/Power BI for analysis and visualization.

The cleaned dataset provides a more reliable foundation for generating meaningful business insights.



Summary:

The customer dataset was successfully cleaned and prepared using Power Query. The major data-quality problems, including missing values, duplicate records, inconsistent entries, formatting issues, and incorrect data types, were addressed.

The transformation process improved the overall quality and consistency of the dataset and made it ready for further analysis.

After completing the cleaning process, all 8 columns contained 100% valid values with no errors or empty entries, as confirmed using the Column Quality feature in Power Query.

Conclusion:

The customer dataset was successfully cleaned and prepared using Microsoft Power Query. Missing values, duplicate records, inconsistent entries, text-formatting issues, and data-type problems were addressed during the transformation process.

The resulting dataset is more accurate, consistent, and suitable for analysis. Data cleaning is an important step because the quality of business decisions depends heavily on the quality of the underlying data.

By using Power Query, the company can create a repeatable and efficient data-preparation process. The cleaned dataset can now be used for customer analysis, reporting, dashboard creation, and strategic business decision-making.

Final Outcome

Before Cleaning:

The dataset contained missing values, duplicate records, inconsistent entries, and formatting/data-type issues.

After Cleaning:

The dataset was standardized, duplicates were removed, missing values were addressed, data types were corrected, and the data was prepared for reliable analysis.

Overall Result:

The cleaned customer dataset provides a stronger and more reliable foundation for business analysis and decision-making.

Business Insights:

- Cleaning customer data provides several important business benefits and enables more reliable insights.
- **Customer Understanding**
- A clean dataset allows the company to accurately understand its customer base. Customer characteristics can be analyzed without duplicate or inconsistent records affecting the results.
- **Improved Customer Segmentation**
- Standardized customer information makes it easier to divide customers into meaningful groups based on attributes such as age, location, purchasing behavior, or customer category.
- **More Accurate Sales Analysis**
- Removing duplicate records prevents customers or transactions from being counted more than once. This helps ensure that sales-related calculations and reports are more accurate.
- **Better Decision-Making**
- Reliable data allows management to make decisions based on accurate information rather than incomplete or inconsistent records.
- **Improved Customer Service**
- A clean customer database can help the company maintain more accurate customer information. This can support better communication, customer management, and service delivery.
- **Identification of Data-Quality Problems**
- The cleaning process also highlights areas where data collection can be improved. For example, recurring missing values or inconsistent entries may indicate that stronger data-entry rules are required.

- **Better Reporting and Visualization**
- Clean and standardized data produces more reliable charts, dashboards, and reports. This allows managers to identify trends and patterns more effectively.